$$\alpha = 4\alpha' \frac{\tanh(\lambda/2)}{\sinh(\lambda n)} \left(\cosh\left(\left(q-\tfrac{1}{2}\right)\lambda\right) \sinh\left(\left(n-\left(\frac{p+q-1}{2}\right)\right)\lambda\right) + \cosh\left(\left(n-\left(p-\tfrac{1}{2}\right)\right)\lambda\right) \sinh\left(\frac{p+q-1}{2}\lambda\right) \right)$$

$$\cosh\left(\left(q-\tfrac{1}{2}\right)\lambda\right) \sinh\left(\left(n-\left(\frac{p+q-1}{2}\right)\right)\lambda\right) + \cosh\left(\left(n-\left(p-\tfrac{1}{2}\right)\right)\lambda\right) \sinh\left(\frac{p+q-1}{2}\lambda\right)$$

$$= \cosh\left(\left(q-\tfrac{1}{2}\right)\lambda\right) \left[\sinh(n\lambda)\cosh\left(\frac{p+q-1}{2}\lambda\right) - \sinh\left(\left(\frac{p+q-1}{2}\right)\lambda\right)\cosh(n\lambda) \right]$$

$$+ \sinh\left(\frac{p+q-1}{2}\lambda\right) \left[\cosh(n\lambda)\cosh\left(\left(p-\tfrac{1}{2}\right)\lambda\right) - \sinh(n\lambda)\sinh\left(\left(p-\tfrac{1}{2}\right)\lambda\right) \right]$$

expand

$$= \sinh(n\lambda) \left[\cosh\left(\left(q-\tfrac{1}{2}\right)\lambda\right)\cosh\left(\frac{p+q-1}{2}\lambda\right) - \sinh\left(\frac{p+q-1}{2}\lambda\right)\sinh\left(\left(p-\tfrac{1}{2}\right)\lambda\right) \right]$$

$$+ \cosh(n\lambda) \left[\sinh\left(\frac{p+q-1}{2}\lambda\right)\cosh\left(\left(p-\tfrac{1}{2}\right)\lambda\right) - \cosh\left(\left(q-\tfrac{1}{2}\right)\lambda\right)\sinh\left(\left(\frac{p+q-1}{2}\right)\lambda\right) \right]$$

$$= \sinh(n\lambda) \left[\cosh\left(\frac{q-p}{2}\lambda\right) + \frac{1}{2}\left[\cosh\left(\frac{3q+p-2}{2}\lambda\right) - \cosh\left(\frac{3p+q-2}{2}\lambda\right) \right] \right]$$

$$+ \cosh(n\lambda) \left[\sinh\left(\frac{p+q-1}{2}\lambda\right) \left[\cosh\left(\left(p-\tfrac{1}{2}\right)\lambda\right) - \cosh\left(\left(q-\tfrac{1}{2}\right)\lambda\right) \right] \right]$$

$$= \sinh(n\lambda)\left[\cosh\left(\frac{q-p}{2}\lambda\right) + \frac{1}{2}\left[\cosh\left(\frac{3q+p-2}{2}\lambda\right) - \cosh\left(\frac{3p+q-2}{2}\lambda\right)\right]\right]$$

$$+ \cosh(n\lambda)\left[\frac{1}{2}\left[\sinh\left(\left(\frac{3p+q-2}{2}\right)\lambda\right) + \sinh\left((q-p)\lambda\right)\right] - \frac{1}{2}\left[\sinh\left(\frac{3q+p-2}{2}\lambda\right) + \sinh\left((p-q)\lambda\right)\right]\right]$$

$$= \sinh(n\lambda)\cosh\left(\frac{q-p}{2}\lambda\right) + \cosh(n\lambda)\left[\sinh\left((q-p)\lambda\right) - \sinh\left((p-q)\lambda\right)\right]$$

$$+ \frac{1}{2}\sinh(n\lambda)\cosh\left(\frac{3q+p-2}{2}\lambda\right) - \frac{1}{2}\sinh(n\lambda)\cosh\left(\frac{3p+q-2}{2}\lambda\right)$$

$$+ \frac{1}{2}\cosh(n\lambda)\sinh\left(\left(\frac{3p+q-2}{2}\right)\lambda\right) - \frac{1}{2}\cosh(n\lambda)\sinh\left(\frac{3q+p-2}{2}\lambda\right)$$

$$= \sinh(n\lambda)\cosh\left(\frac{q-p}{2}\lambda\right) + \cosh(n\lambda)\left[\sinh\left((q-p)\lambda\right) - \sinh\left((p-q)\lambda\right)\right]$$

$$+ \frac{1}{2}\sinh(n\lambda)\cosh\left(\frac{3q+p-2}{2}\lambda\right) - \frac{1}{2}\cosh(n\lambda)\sinh\left(\frac{3q+p-2}{2}\lambda\right)$$

$$- \frac{1}{2}\sinh(n\lambda)\cosh\left(\frac{3p+q-2}{2}\lambda\right) + \frac{1}{2}\cosh(n\lambda)\sinh\left(\left(\frac{3p+q-2}{2}\right)\lambda\right)$$

check!

$$= \sinh(n\lambda)\cosh\left(\frac{q-p}{2}\lambda\right) + \cosh(n\lambda)\left[\sinh\left((q-p)\lambda\right) - \sinh\left((p-q)\lambda\right)\right]$$

$$+ \frac{1}{2}\sinh\left(\left(n - \left(\frac{3q+p-2}{2}\right)\right)\lambda\right) - \frac{1}{2}\sinh\left(\left(n - \left(\frac{3p+q-2}{2}\right)\right)\lambda\right)$$

$$=$$